

CTCC Documentation To-Do

This document tracks which method documentation (.tex) files we need. Standard for each doc: concept + math/model + relevant code.

1 Inventory: CTCC methods (scripts)

Scope: scripts/ (25 files) and ctcc.py (project root). Other code not yet inventoried: server/app (API). Excludes sensitivity_analysis.py, oat_analysis.py, sensitivity_utils.py, and related analysis scripts.

1. ctcc.py
2. weighted_miles.py
3. build_costs.py
4. insurance_costs.py
5. row_costs.py
6. environmental_mitigation.py
7. delay_costs.py
8. wildfire_costs.py
9. outage_costs.py
10. congestion_curtailement_reduction.py
11. energy_losses.py
12. emissions.py
13. line_loss_costs.py
14. oandm.py
15. bcr_calculator.py
16. csv_output_manager.py

17. `financial_utils.py`
18. `calculation_utils.py`
19. `constants.py`
20. `json_loaders.py`
21. `json_output_manager.py`
22. `yaml_loaders.py`
23. `smart_loaders.py`
24. `smart_output.py`
25. `path_config.py`
26. `revenue.py`

2 Mapping: script $\rightarrow$ documentation .tex

- `ctcc.py` $\rightarrow$ `ctcc_documentation.tex`
- `weighted_miles.py` $\rightarrow$ `weighted_miles_documentation.tex`
- `build_costs.py` $\rightarrow$ `build_costs_documentation.tex`
- `insurance_costs.py` $\rightarrow$ `insurance_costs_documentation.tex`
- `row_costs.py` $\rightarrow$ `row_costs_documentation.tex`
- `environmental_mitigation.py` $\rightarrow$ `environmental_mitigation_documentation.tex`
- `delay_costs.py` $\rightarrow$ `delay_costs_documentation.tex`
- `wildfire_costs.py` $\rightarrow$ `wildfire_costs_documentation.tex`
- `outage_costs.py` $\rightarrow$ `outage_costs_documentation.tex`
- `congestion_curtailement_reduction.py` $\rightarrow$ `congestion_curtailement_reduction_documentation.tex`
- `energy_losses.py` $\rightarrow$ `energy_losses_documentation.tex`
- `emissions.py` $\rightarrow$ `emissions_documentation.tex`
- `line_loss_costs.py` $\rightarrow$ `line_loss_costs_documentation.tex`
- `oandm.py` $\rightarrow$ `oandm_documentation.tex`
- `bcr_calculator.py` $\rightarrow$ `bcr_calculator_documentation.tex`

- `csv_output_manager.py` → `csv_output_manager_documentation.tex`
- `financial_utils.py` → `utilities_documentation.tex`
- `calculation_utils.py` → (in `utilities_documentation.tex`?)
- `constants.py` → (in `utilities_documentation.tex`?)
- `json_loaders.py` → `json_loaders_documentation.tex`
- `json_output_manager.py` → `json_output_manager_documentation.tex`
- `yaml_loaders.py` → `yaml_configuration_guide.tex` (partial?)
- `smart_loaders.py` → `smart_loaders_documentation.tex`
- `smart_output.py` → `smart_output_documentation.tex`
- `path_config.py` → `path_config_documentation.tex`
- `revenue.py` → `revenue_documentation.tex`

3 Plan A: Slates (fill from scratch)

Scope: Docs that are placeholders only: `ctcc_documentation.tex`, `weighted_miles_documentation.tex`, `json_loaders_documentation.tex`, `json_output_manager_documentation.tex`, `smart_loaders_documentation.tex`, `smart_output_documentation.tex`, `path_config_documentation.tex`, `revenue_documentation.tex`.

Goal: Turn each slate into a complete doc (concept + math/model + relevant code). One doc per iteration; same steps each time. Reference structure: `congestion_curtailement_reduction_documentation.tex`

3.1 Process (per slate)

1. Open the script and its `.tex` side by side.
2. **Introduction:** Replace the placeholder with one or two sentences on what the script does and its role in CTCC.
3. **Method overview:** Purpose: replace with a short numbered list of what the script actually does (trace `main()` and key calls). Calculation approach: if there is real logic/formulas, write equations to match the code; if it is I/O or dispatch only, say so in one sentence. Inputs/Outputs: fill from the script (what it loads, what it writes; env vars if relevant). Integration: list the scripts it imports from and who imports it (or “used by all calculation scripts when...”).
4. **Key functions:** For each important function, add a subsection: name/signature, parameters, return value, and a short “Key logic” `lstlisting` snippet from the current script.

5. **Example:** Replace with a minimal example (e.g. env vars + one call, or one input/output snippet).
6. Compile the `.tex` and skim the PDF; mark that slate done in the tracking list.

3.2 To-dos (one per slate)

1. Pass (slate): `ctcc_documentation.tex`
2. Pass (slate): `weighted_miles_documentation.tex`
3. Pass (slate): `json_loaders_documentation.tex`
4. Pass (slate): `json_output_manager_documentation.tex`
5. Pass (slate): `smart_loaders_documentation.tex`
6. Pass (slate): `smart_output_documentation.tex`
7. Pass (slate): `path_config_documentation.tex`
8. Pass (slate): `revenue_documentation.tex`

4 Plan B: Existing docs (update / verify)

Scope: Docs that already have content: `build_costs_documentation.tex`, `insurance_costs_documentation.tex`, `row_costs_documentation.tex`, `environmental_mitigation_documentation.tex`, `delay_costs_documentation.tex`, `wildfire_costs_documentation.tex`, `outage_costs_documentation.tex`, `congestion_curtailement_reduction_costs_documentation.tex`, `energy_losses_documentation.tex`, `emissions_documentation.tex`, `line_loss_costs_documentation.tex`, `oandm_documentation.tex`, `bcr_calculator_documentation.tex`, `csv_output_manager_documentation.tex`, `utilities_documentation.tex`, `yaml_configuration_guide.tex`. Also any doc that covers calculation utils or constants (e.g. inside `utilities_documentation.tex`).

Goal: Bring each doc in line with the current script: fix concept drift, refresh math, and refresh code. One doc per iteration; same steps each time.

4.1 Process (per existing doc)

1. Open the script and its `.tex` side by side.
2. **Concept check:** Introduction: does it still describe what the script does? Update if the script's role or steps have changed. Purpose: do the bullets match the script's current flow (e.g. new steps, reordering, removed logic)? Inputs/Outputs: align with current loaders and output usage (env, file paths, keys). Integration: update the "integrates with" list to match current imports and callers.
3. **Math check:** Calculation approach: do equations, variable names, and edge cases (e.g. reconductoring, zero credits) match the code? Update or add equations from the script/comments.

4. **Code check:** Key functions: do the listed functions still exist with the same role? Add any new important functions; remove or rewrite subsections for renamed/removed logic. Snippets: replace every `\lstlisting` with the corresponding current code (or a short excerpt). Fix any outdated variable names or logic.
5. **Example:** If there is an example, check it still runs or matches current behavior; update or add one if missing.
6. Compile the `.tex` and skim the PDF; mark that doc done in the tracking list.

4.2 To-dos (one per existing doc)

1. Pass (update): `build_costs_documentation.tex`
2. Pass (update): `insurance_costs_documentation.tex`
3. Pass (update): `row_costs_documentation.tex`
4. Pass (update): `environmental_mitigation_documentation.tex`
5. Pass (update): `delay_costs_documentation.tex`
6. Pass (update): `wildfire_costs_documentation.tex`
7. Pass (update): `outage_costs_documentation.tex`
8. Pass (update): `congestion_curtailment_reduction_documentation.tex`
9. Pass (update): `energy_losses_documentation.tex`
10. Pass (update): `emissions_documentation.tex`
11. Pass (update): `line_loss_costs_documentation.tex`
12. Pass (update): `oandm_documentation.tex`
13. Pass (update): `bcr_calculator_documentation.tex`
14. Pass (update): `csv_output_manager_documentation.tex`
15. Pass (update): `utilities_documentation.tex`
16. Pass (update): `yaml_configuration_guide.tex`